Check for updates

Anaesthesia 2026 https://doi.org/10.1111/anae.70284

Review Article

Management of major thoracic trauma: a narrative review

Navniel Kaur, Sophie Lewis, Hollie Jenkinson, Christopher Keegan, Nikhil Misra, Simon Neequaye, Simon Scott, Tim Astles and **Ben Morton**

Summary

Introduction Traumatic thoracic injuries are a leading cause of morbidity and mortality worldwide, not only as a direct consequence of the injury itself, but also due to development of secondary insults such as infection; acute kidney injury; acute respiratory distress syndrome; and shock. The thorax houses vital structures, and even seemingly trivial trauma may result in catastrophic damage particularly in children or older patients. This review aims to highlight important considerations in diagnosis and management of major thoracic injuries and the crucial role of appropriate supportive interventions such as airway control, ventilation strategies and optimal pain management to aid in recovery.

Methods We undertook a narrative review of the contemporary literature. We synthesised evidence related to mechanisms of thoracic trauma and initial assessment and diagnosis of major thoracic injuries. We also highlight key considerations in chest wall, pulmonary, cardiac and diaphragmatic injuries; supportive management strategies for patients with thoracic trauma; and special considerations in older patients.

Results Thoracic injuries are broadly classified by the mechanism of trauma (blunt, penetrating or blast) or by the anatomical structure involved. Initial assessment and diagnosis follows general trauma principles. Depending on the injuries sustained, surgical intervention may be indicated. Equally important is appropriate supportive management. Older patients are often overlooked as a high-risk group and a lower threshold for specialised trauma care or critical care services is recommended in this cohort.

Discussion A multidisciplinary approach is key in management of patients with thoracic trauma due to their numerous needs, ranging from surgical intervention and critical care support to physiotherapy and rehabilitation.

For full author affiliations, see end of article

Plain Language Summary is available on the journal website.

Correspondence to: Navniel Kaur
Email: navnielk@gmail.com

Accepted: 15 April 2026
Keywords: diagnosis; management; mechanisms; multidisciplinary; thoracic trauma

Bluesky: @benmorton.bsky.social
X: @NavnielKaur

Introduction

Thoracic trauma is defined as injury to the bony chest wall (ribs and sternum) or mediastinal structures due to an external mechanical force [1]. The global incidence of severe thoracic trauma is estimated at approximately 10 million cases per year [2]. Between 71% and 85% of people affected are male [3–6] with reported mortality rates of 9–11% [3, 6, 7].

The thorax is the second most commonly injured body region (46%) in people suffering major trauma [8]. Mortality after thoracic trauma may be immediate due to catastrophic damage to the heart or major vessels, or late due to development of complications, such as respiratory compromise or sepsis. The Abbreviated Injury Scale (AIS) classifies injury to each body region on a six-point scale [9]. Thoracic injuries often score higher [10] as the thorax

© 2026 Association of Anaesthetists.

houses vital structures that, if injured, can cause rapid deterioration [11]. Patients with thoracic trauma and AIS scores ≥5 suffer higher rates of sepsis (16% vs. 7%); respiratory failure (40% vs. 14%); multi-organ failure (42% vs. 19%); and early hospital mortality (22% vs. 5%) compared with patients with thoracic trauma and AIS scores ≤ 1 [11].

We aimed to highlight important considerations for diagnosis of thoracic trauma and the importance of multidisciplinary supportive management, including collaboration between the various clinical teams, to ensure best outcomes.

Methods

We undertook a narrative review of the contemporary literature, drawing on systematic reviews; large observational studies; randomised controlled trials; and national and international guidelines. The literature search was conducted using Google Scholar, PubMed and MEDLINE. Key search terms included: `thoracic´; `trauma´; `pulmonary´; `chest wall´; `cardiac´; `diaphragm´; and `supportive management´. We took a multidisciplinary approach with specialty teams responsible for individual sections of the review. These were: anaesthesia (NK, BM and TA); critical care (SL, BM and TA); general surgery (HJ and NM); orthopaedics (SS); vascular surgery (SN); and interventional radiology (CK). The specialist teams drafted individual sections before synthesis of the results.

Results

Mechanisms and classification of thoracic trauma

Thoracic injuries are classified broadly based on the mechanism of trauma: blunt; penetrating; or blast [1, 12]. Blunt injuries cause impact without disrupting the skin barrier and are often accompanied by skeletal fractures. Road traffic collisions are the most common cause of blunt thoracic trauma [13, 14]; other causes include falls from a height, physical assault and sports injuries. Penetrating injuries occur due to laceration of the anatomic structures of the chest cavity and may be ballistic (e.g. gunshot) or non-ballistic (e.g. blade) [10, 12]. The kinetic energy, force and characteristics of the weapon determine the severity of the trauma [15]. Velocity is used to classify penetrating wounds as high-energy (> 600 $m.s^{-1}$); intermediate-energy (340–600 $m.s^{-1}$); or low-energy (< 340 $m.s^{-1}$) [16, 17]. Blunt injuries have higher mortality rates than penetrating trauma, but the latter are more likely to require surgical intervention [6]. Although figures vary dependent on economic, social and hospital location factors, blunt-force thoracic trauma is more common than penetrating injury [3, 8].

Blast injuries comprise a special category as the mechanism of injury combines both blunt and penetrating insults, depending on the phase of the blast trauma. For example, the primary phase of injury typically causes blunt injuries due to shockwaves generated from the blast, while the secondary phase may result in penetrating injuries due to shrapnel or debris [12, 17, 18].

Injuries may also be described based on the anatomical structure involved, for example: chest wall; pulmonary; cardiac and vascular; diaphragmatic; and tracheobronchial. The frequency of injuries as per anatomical region varies in different settings but rib fractures occur most frequently (33–70%), followed by lung parenchymal injuries; cardiac injuries occur less frequently (3.3–5.0%) [3, 19, 20]. The severity of thoracic injuries is graded on a scale of 1–4 (Table 1) [21]. Understanding the classification, mechanism and severity of injuries assists in informing clinical management.

Initial assessment and diagnosis

Pre-hospital management of patients with thoracic injury should follow standardised principles [22, 23]. Airway protection, maintenance of adequate ventilation, haemorrhage control and management of shock are key. The European Trauma Course (ETC) and Advanced Trauma Life Support (ATLS) are recognised principles for managing major trauma. The ATLS is an internationally recognised system introduced by the American College of Surgeons and focuses on a systematic approach to managing trauma [22]. The European Trauma Course is a newer system which incorporates a more multidisciplinary approach [23]. The ATLS guidelines have limitations particularly in modern trauma centres [24]; however, the guidelines still form the basics of management for several trauma organisations and are relevant for resident doctors or those who do not manage trauma cases routinely.

Both ETC and ATLS principles involve a comprehensive primary survey with an `A to E´ assessment of the patient [22, 23]. In addition, ETC guidelines involve the team leader performing a `5 second round´, a brief initial assessment to identify immediate life-threatening conditions such as complex airway obstruction; massive external haemorrhage; or traumatic cardiac arrest. This is achieved using the `Assessment Triangle´, a tool providing an immediate indication of the severity of the patient's condition based on their social interaction, respiratory effort and skin perfusion [23]. Key components of ETC and ATLS guidelines are to secure venous access rapidly; take appropriate blood samples; and obtain radiological imaging.

© 2026 Association of Anaesthetists.

13652044, 0, Downloaded from https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.70284 by Hacettepe University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1 Classification of thoracic trauma. The two classifications are not equivalent grading systems.

Abbreviated Injury Scale (2015) [9]	
Severity	**Examples**
1 Minor	One rib fracture Myocardial contusion
2 Moderate	Two fractured ribs Pneumomediastinum
3 Serious	Three fractured ribs or a unilateral flail segment Haemothorax (≤ 20% blood loss) Pneumothorax (≤ 50% lung collapse, no tension)
4 Severe	More than five flail ribs Haemothorax (> 20% blood loss) Pneumothorax (> 50% lung collapse, no tension)
5 Critical	Bilateral flail segments Tension pneumothorax Aortic injury haemorrhage confined to the mediastinum
6 Unsurvivable	Traumatic ventricular rupture Aortic injury with haemorrhage not confined to the mediastinum

World Society of Emergency Surgery and American Association for the Surgery of Trauma (2025) [21]			
Severity	**Grade**	**Respiratory function**	**Haemodynamic status**
Minor	1	Normal	Stable
Moderate	2	Respiratory impairment*	Stable
Severe	3	Respiratory distress**	Stable
	4	Any	Unstable

*Decreased lung function without distress.
**Acute diffuse lung injury with potential for severe hypoxaemia.

In patients who are haemodynamically unstable, the extended Focused Assessment with Sonography in Trauma (FAST) scan and/or chest radiographs are recommended for initial diagnosis. In patients with stable injuries, contrast-enhanced computerised tomography (CT) images with 3D reconstruction are recommended [21], and CT angiography is advised to define vascular involvement allowing for appropriate interventions. Bronchoscopy is the gold standard for diagnosis of bronchial and tracheal injuries [21]. According to NHS guidelines, a trauma CT scan should ideally be performed within 30 min of the patient's arrival at hospital [25] and a final report should be available within 1 h of the scan [26].

Thoracic injuries and their management

Chest wall injuries

Rib fractures are caused most commonly by blunt force trauma, especially falls. They adversely affect older patients who often sustain multiple, displaced rib fractures from relatively minor falls [27]. Chest radiographs have low sensitivity in diagnosis of rib fractures, with up to 50% of fractures missed [28]. A chest CT scan is indicated in cases of major trauma; where there is high suspicion of a fracture despite a negative chest radiograph; or in high-risk groups such as patients aged > 65 y [28].

A clinical flail chest occurs when a segment of chest wall detaches due to multiple (three or more) rib fractures in two or more places. This results in paradoxical movement of the flail segment which disrupts normal respiratory mechanics and impairs effective ventilation. A radiological flail is a potential flail identified during review of a chest CT confirming three or more ribs fractured in more than one place, without any apparent clinical features seen in a complete flail (Fig. 1). Sternal fractures are commonly caused by anterior chest wall trauma, especially from road traffic collisions [29] or falls from a low level which appear to be trivial in older patients [30].

Chest wall injuries require prompt management to prevent complications such as atelectasis and pneumonia. Uncomplicated, minimally displaced chest wall injuries are treated with supplemental oxygen, multimodal analgesia and early mobilisation/physiotherapy. Multiple displaced rib fractures, bilateral displaced rib fractures and flail segments are more serious injuries, and patients may require ventilatory support.

© 2026 Association of Anaesthetists.

13652044, 0, Downloaded from https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.70284 by Hacettepe University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

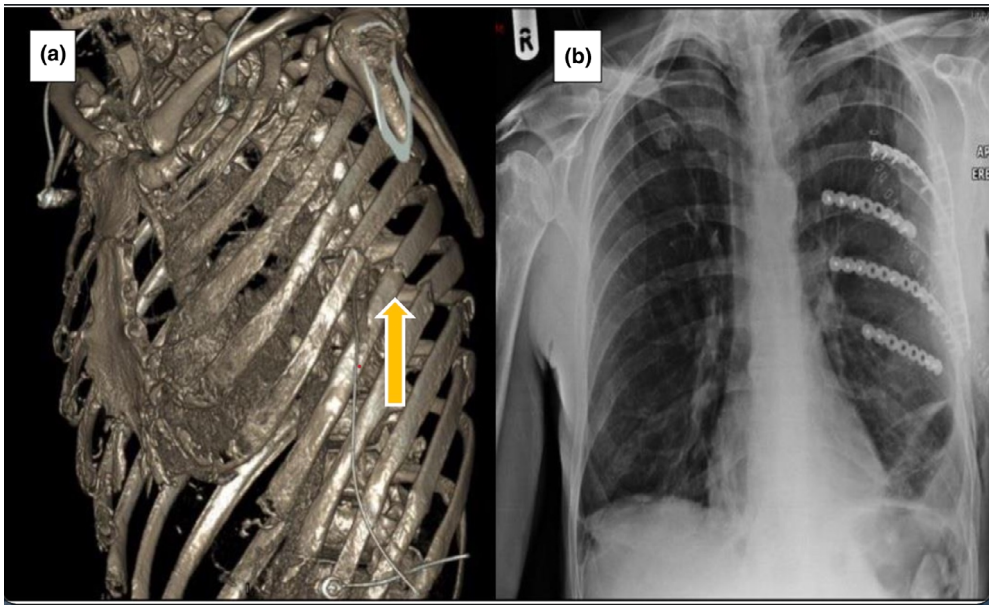


Figure 1 Flail chest and surgical fixation of rib fractures. (a) 3D reconstruction of CT thorax showing flail chest; (b) chest radiograph showing rib plates following surgical fixation.

Surgical fixation of severe chest wall injuries may be required to provide stability, reduce pain and promote healing. Indications for rib fracture fixation are highlighted in Box 1 [31]. Ideally, fixation should be done within 48–72 h of injury. If this is not possible, it should be done as soon as is feasible, preferably within 3–7 days of injury [21]. Early surgical intervention aids recovery by reducing: inflammation; haematoma formation; development of an empyema; and deformities. Timely fixation also prevents early callous formation which would make later surgical repair more difficult. In patients who are haemodynamically unstable, surgical fixation of rib fractures is not recommended [21]. Relative contraindications to surgical rib fixation include active malignancy; severe cardiac or pulmonary comorbidities; and advanced age [21].

The surgical fixation approach is defined pre-operatively by the pattern of the rib fractures. Ribs 3–8 are the fractures fixed most commonly [32–38]. Rib fixation is not usually attempted for injuries of ribs 1, 2, 11 and 12 as these are difficult to access and less critical to maintain adequate ventilation. An additional consideration is that a rib length > 2.5 cm on each side of the fracture is required to affix a plate [31]. Surgeons attempt to fix all fractures through a single incision where possible to reduce postoperative pain and infection risk. Whilst sternal fractures are not generally treated surgically, displaced fractures may be repaired to reduce pain and risk of non-union.

Box 1 Indications for surgical rib fixation [31].

- Flail chest
- Multiple (≥ three) ipsilateral severely displaced rib fractures
- Multiple (≥ three) ipsilateral displaced rib fractures in ribs 3–10 in combination with one of:
  - respiratory failure (despite mechanical ventilation);
  - weaning failure; or
  - ≥ two pulmonary derangements in patients whose lungs are not mechanically ventilated despite loco-regional anaesthesia and multimodal analgesia including: respiratory rate > 20 breaths.min^{-1}; incentive spirometry < 50% predicted; numeric pain score > 5/10; or poor cough
- Chest wall deformity significantly affecting lung function or showing mechanical instability on palpation
- Severe pain unresponsive to other treatments

Pulmonary injuries

Pulmonary trauma may cause parenchymal injuries (contusions, lacerations, haematomata or herniation); pneumothorax; or haemothorax [39]. Lung contusions

© 2026 Association of Anaesthetists.

13652044, 0, Downloaded from https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.70284 by Hacettepe University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

occur in up to 19–31% of patients suffering thoracic trauma [40, 41], with road traffic collisions and falls the leading causes [40, 42]. Minor contusions are usually asymptomatic, and management is generally supportive. Severe lesions may cause reduced lung compliance and impaired gas exchange, leading to respiratory distress [42–44]. Acute respiratory distress syndrome and pneumonia are the most frequent complications of pulmonary contusions, with risk proportional to the severity of injury [42, 43, 45–47].

Pulmonary lacerations occur most commonly following penetrating injuries. The resulting laceration cavity may fill with air (pneumatocele); blood (haematoma or haematocele); or both (haematopneumatocele) [39]. These injuries usually heal spontaneously over weeks to months. Complications occur infrequently but include: abscess formation; laceration enlargement via ball-valve effect; bronchopleural fistula with a persistent air leak [48]; or vascular complications causing an arteriovenous fistula or pseudoaneurysm [49].

The increased use of investigative CT scans has led to a rise in diagnosis of pneumothoraces not apparent on chest radiographs (occult pneumothorax) [50]. Asymptomatic and small pneumothoraces are treated conservatively and monitored with serial radiographs for at least 24 h [21, 51]. Success rates of >80–85% have been found with conservative management [50, 52]. An intercostal drain is recommended for asymptomatic patients who have > 20% pneumothorax on chest radiograph (measured on a supine radiograph) or a pneumothorax size of > 35 mm on CT scan [51]. The measurement is performed on an axial view of a chest CT, with a line drawn in the largest air pocket radial or perpendicular to the chest wall. For symptomatic cases, thoracostomy is performed and an intercostal drain is inserted [53]. In patients with an occult pneumothorax requiring short-term positive pressure ventilation of their lungs, routine intercostal drain insertion is not recommended. However, intercostal drain placement is recommended if positive pressure ventilation anticipated to last > 4 days [54]. Small diameter intercostal drains (20-Fr and smaller) or 14-Fr pigtail catheters are recommended in adults, instead of traditional larger sized drains; larger drains may cause significant pain via compression of the neurovascular bundle along the lower edge of the rib [55, 56].

Open sucking chest wounds require a chest seal, debridement and closure of the wound. The persistent presence of air in the pleural space despite drainage suggests leakage, tracheobronchial injury, bronchopleural fistula or poor lung re-expansion due to underlying parenchymal injury or chronic disease, and surgery may be indicated [57].

Haemothorax frequently occurs together with a pneumothorax, usually at the time of injury. However, in up to one-third of patients haemothorax may be delayed, especially with posterior rib fractures or displaced rib fractures [58]. Thoracostomy and intercostal drain insertion is the choice of management. Drainage is recommended in stable patients with a haemothorax of ≥ 500 ml or in unstable patients irrespective of the volume [21]. Conservative management may be used in stable patients with haemothorax volumes of < 300 ml [21]. Clinical features should influence the management approach for both haemo- and pneumothoraces, irrespective of volumes [21]. Antibiotic prophylaxis is not routinely recommended for intercostal drain insertion, unless penetrating injuries are present [59].

An emergency thoracotomy may be required in cases of massive or ongoing haemorrhage. Indications include: 1500 ml of blood drained in 24 h; 300–500 ml within 2–4 h after drain insertion; or the need for repeated blood transfusions to maintain haemodynamic stability [60, 61]. Surgical exploration allows control at the source of bleeding and the opportunity to evacuate intrathoracic blood [61].

Cardiac injuries

Myocardial contusion after blunt thoracic trauma is most often caused by deceleration injuries. Associated complications range from minor non-specific arrhythmias to life-threatening myocardial dysfunction and/or pericardial tamponade. Diagnosis is challenging due to the lack of specific tests [62] with chest radiographs and CT of limited value [63]. If cardiac injury is suspected, a 12-lead ECG and cardiac enzymes should be measured and a focused cardiac ultrasound performed [1, 64]. A patient with suspected myocardial contusion should have continuous cardiac monitoring for 24–48 h due to the risk of life-threatening arrythmias or cardiac failure. Depending on concomitant injuries, and the presence and type of ECG changes, continuous telemetry may be performed on a specialised cardiac ward or the ICU [65]. Echocardiography is recommended in patients with thoracic trauma with any suspicion of: cardiac involvement or pericardial tamponade; haemodynamic instability despite ongoing resuscitation; abnormal ECG or deranged cardiac markers; unexplained hypotension; or cardiac arrest or peri-arrest following thoracic trauma [66].

Cardiac tamponade is life-threatening and requires immediate intervention. Tamponade is caused commonly

© 2026 Association of Anaesthetists.

13652044, 0, Downloaded from https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.70284 by Hacettepe University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

by penetrating injuries and leads to haemodynamic instability by compression of the heart impairing filling, leading to circulatory shock, cardiac arrest and death [67]. The extended FAST approach may be used to rapidly diagnose cardiac tamponade especially if haemodynamic instability is present [68]. Definitive management of traumatic cardiac tamponade includes surgical drainage via a thoracotomy or subxiphoid pericardial approach [69]. Percutaneous drainage is recommended as a temporary stabilising measure, especially in low-resource or prehospital settings, and is usually inadequate in cardiac injuries or active haemorrhage [70].

Although less common than blunt injuries, penetrating cardiac injuries have a greater risk of mortality [71] and early diagnosis and intervention is vital [72]. If a patient is haemodynamically stable, appropriate surgical intervention should be performed as soon as possible in the operating theatre in the form of a sternotomy [73]. In the event of peri-arrest or cardiac arrest, surgical intervention in the form of a resuscitative thoracotomy, may be necessary in the emergency department [74]. The European Resuscitation Council recommends the time from loss of signs of life to commencement of a resuscitative thoracotomy should not be > 15 min after penetrating trauma [75].

Indications for an emergency department resuscitative thoracotomy are: peri-arrest state and unable to tolerate transfer to an operating theatre; established cardiac arrest with penetrating chest trauma with < 15 min of cardiopulmonary resuscitation; or established cardiac arrest with blunt trauma with < 10 min of cardiopulmonary resuscitation (pulseless but with other signs of life such as pupillary response) [74]. Contraindications for an emergency department resuscitative thoracotomy are: established cardiac arrest with cardiopulmonary resuscitation for longer than the times stated previously; and the presence of other major non-survivable injuries [76].

When performing an emergency department resuscitative thoracotomy, a left anterolateral approach is advised. When the thoracic cavity is accessed, the pericardium should be opened with a vertical incision, taking care to avoid injuring the phrenic nerve. The anterolateral incision can be extended to a clamshell thoracotomy for increased exposure [73]. When performed correctly and in a timely manner, resuscitative thoracotomy can improve patient outcomes. A retrospective review of prehospital resuscitative thoracotomy for traumatic cardiac arrests showed a survival to hospital discharge rate of 5.0%, with a good neurological status recorded in 76.6% of survivors [77]. Patient selection and time from arrest is crucial; no survivors were identified in patients who suffered cardiac arrest for > 5 min after exsanguination. For patients with traumatic cardiac arrest due to cardiac tamponade, survival rates are higher if the thoracotomy is performed within 15 min of arrest [77].

Great vessel injuries

Damage to the aorta and great vessels may occur with blunt or penetrating trauma. Blunt thoracic aortic injury is rare but associated with a very high mortality. The mechanism of injury relates to rapid deceleration of the aortic arch in opposition to the descending thoracic aorta, which is anchored anatomically by the intercostal arteries. The 2025 guidelines from the European Society for Vascular Surgery provide a simplified classification of aortic injury (Fig. 2) [78]. Grade 3 injuries involve total vessel wall transection, with only 25% survival to hospital presentation due to bleeding complications and associated injuries [79]. Penetrating thoracic aortic injury is less frequent than blunt injury and associated with worse outcomes [80], with a younger, predominantly male cohort affected. Patients are less likely to survive to hospital and much more likely to have associated oesophageal, cardiac and diaphragmatic injury that require open repair [80].

Thoracic endovascular aortic repair (TEVAR) is preferred to open surgery in patients with suitable anatomy, regardless of age. Figure 3 illustrates a case of traumatic aortic injury before and after TEVAR. Open surgical repair is associated with worse outcomes compared with TEVAR, with observed mortality of open surgical repair double that of TEVAR (15.6% vs. 7.6%); higher rate of paraplegia (5.6% vs. 0.0%); and higher rate of stroke (5.3% vs. 0.8%) [81]. Due to the risk of free rupture, TEVAR should be considered in cases with significant or grade 2 blunt thoracic aortic injury [78]. Emergency TEVAR should be prioritised in patients with free rupture of a blunt thoracic aortic injury or a large peri-aortic haematoma (≥ 15 mm). In blunt thoracic aortic injury without large haematoma, delayed intervention

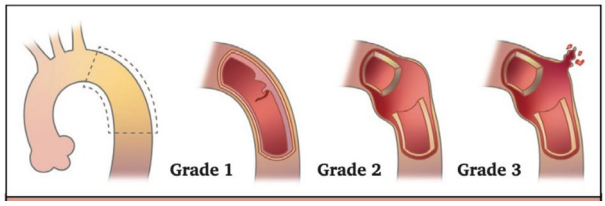


Figure 2 The European Society for Vascular Surgery classification of traumatic thoracic injury. Grade 1, injury confined to the intima or vessel wall, with normal external wall contour; grade 2, external wall disruption with contained haemorrhage; grade 3, complete wall transection with free rupture. Image obtained from [78].

© 2026 Association of Anaesthetists.

13652044, 0, Downloaded from https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.70284 by Hacettepe University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

should be considered to prioritise treatment if the patient has other life-threatening injuries [82]. The balance of risk of immediate vs. delayed TEVAR is critical. Delaying interventions when possible is associated with a lower risk of surgical complications and death [83].

Diaphragmatic injuries

Traumatic diaphragmatic injuries are uncommon. Penetrating diaphragmatic injuries are usually smaller and more difficult to detect compared with blunt injuries which are usually larger and easier to diagnose but associated with higher mortality rate due to presence of co-existing injuries [84]. Diaphragmatic injury is more common on the left side, possibly due to the protective cushion of the liver on the right. An alternative explanation in penetrating trauma is that assailants are more likely to be right-handed, thus inflicting injuries on the left side of the body [85]. A left-sided diaphragmatic injury is often associated with herniation of the spleen and stomach into the thoracic cavity [86]. Diagnosis is challenging and frequently only identified intra-operatively during surgery for co-existing injuries. Although a CT scan is more reliable than a chest radiograph, surgical exploration should be considered if traumatic diaphragmatic injuries are suspected, irrespective of negative radiological findings [87].

If a patient sustains a penetrating injury to the abdomen or chest from T4-T12 dermatomes anteriorly and L3 area posteriorly, there should be a high index of suspicion for diaphragmatic injury. Similarly, if a patient has suffered trauma and has respiratory distress with signs of mediastinal shift or lung displacement or if bowel sounds are auscultated in the chest, a diaphragmatic injury should be suspected [85]. Delayed diagnosis increases the risk of strangulation of intra-abdominal organs through the resultant hernia, which significantly increases mortality risk [87]. Surgical repair is indicated as soon as possible [21], which may be via laparoscopy, thoracoscopy or a laparotomy depending on the pattern and surgical requirements for other injuries. A tension pneumothorax may develop with a laparoscopic approach due to the abdominal insufflation, necessitating intercostal drain insertion or conversion to an open approach [21]. Once identified and the intra-abdominal organs have been returned to their correct anatomical position, the lacerations are repaired with non-absorbable sutures and an intercostal drain is always inserted [21, 88]. If the diaphragmatic injury is too large to be repaired with sutures, a non-absorbable mesh is used for repair with insertion of an intercostal drain [85]. Complications related to repair include: formation of a subphrenic abscess; empyema; wound dehiscence; and phrenic nerve injury leading to diaphragmatic paralysis [21].

Clinical management

The goal of management in patients with thoracic injuries is to optimise physiological conditions and provide appropriate organ system support. Areas of specific interest include: oxygenation; fluid management; and adequate analgesia. The scope of interventions is broad, ranging from a patient with isolated rib fractures with high analgesic requirements to those who are multiply injured and have complex multidisciplinary needs.

Pain management is an important consideration for all patients with thoracic injuries. Timely assessment of appropriate pain scores is necessary and multimodal analgesia is utilised. Figure 4 summarises an example of an analgesic pathway for patients with rib fractures. Alternative

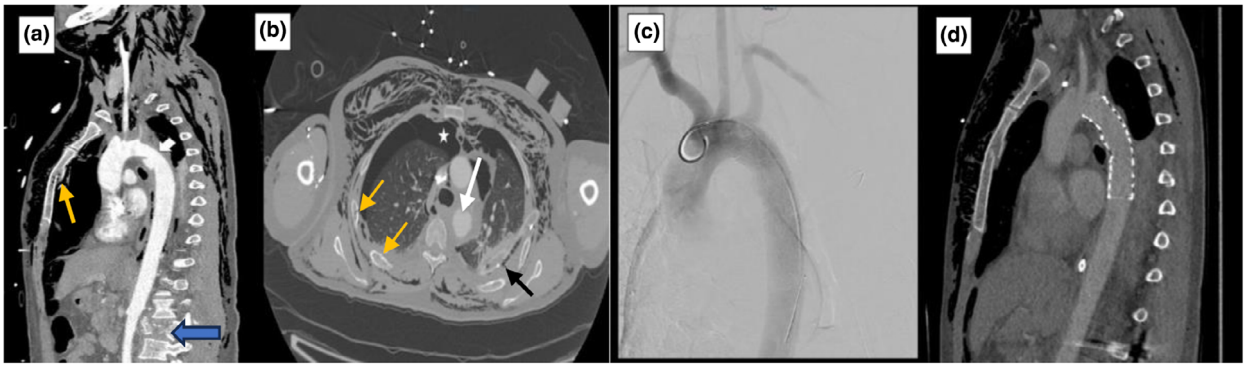


Figure 3 Aortic injury. (a) Multiplanar reformat contrast enhanced CT showing aortic transection at the isthmus (white arrow), sternal fracture (orange arrow), L2 burst fracture (blue arrow) and extensive surgical emphysema tracking into the neck and head; (b) Axial contrast enhanced CT showing aortic transection (white arrow), pneumothorax (white star), intercostal drain (black arrow) and rib fractures (orange arrows); (c) digital subtraction angiogram and (d) contrast enhanced CT post thoracic endovascular aortic repair showing adequate coverage of the defect and patent arch vessels.

© 2026 Association of Anaesthetists.

13652044, 0, Downloaded from https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.70284 by Hacettepe University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

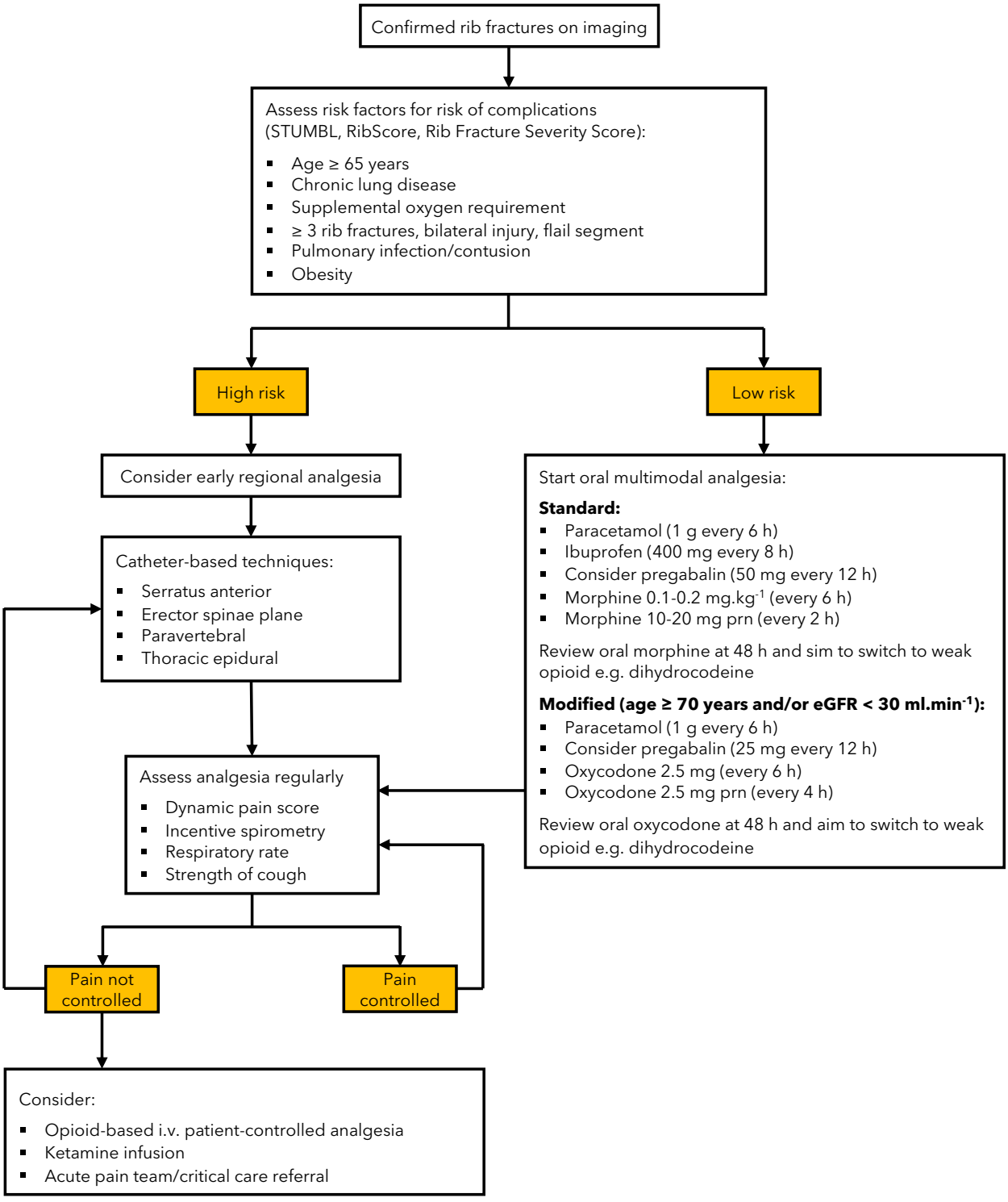


Figure 4 Pain management strategies in patients with rib fractures. Adapted from Sheffield Teaching Hospitals NHS Foundation Trust chest injury management pathway.

scoring systems focus on development of complications and prognosis. The Study of the Management of Blunt Chest Wall Trauma (STUMBL) Score [89] is used to predict the development of complications in blunt thoracic trauma (Table 2). The RibScore uses CT scan findings to predict clinical outcomes such as respiratory failure; pneumonia;

© 2026 Association of Anaesthetists.

13652044, 0, Downloaded from https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.70284 by Hacettepe University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2 The STUdy of the management of BLunt chest wall trauma (STUMBL) score [89].

	Score	
Age	1 point	Each decade
Number of rib fractures	3 points	Each rib fracture*
Pre-injury use of anticoagulants	4 points	Yes
Chronic lung disease	5 points	Yes
Oxygen saturation levels on room air	0 points	≥ 95%
	2 points	94–90%
	4 points	89–85%
	6 points	84–80%
	8 points	79–75%
	10 points	74–70%
Total risk score	**Probability of developing complications****	
0–10	13%	
11–15	29%	
16–20	52%	
21–25	70%	
26–30	80%	
≥ 31	88%	

*Two fractures on one rib = 6 points.
**Composite of in-hospital mortality, morbidity including all pulmonary complications (chest infection, pneumonia, haemothorax, pneumothorax, pleural effusion or empyema); ICU admission; or a prolonged duration of hospital stay (defined as a total hospital stay ≥ 7 days).

tracheostomy; and mortality [90]. The Chest Trauma Score may be useful in situations where CT imaging may not be available [91]. The Rib Fracture Score provides guidance on the need for management in a high-dependency unit and analgesic strategies [92, 93].

Regional techniques are used frequently as adjuncts to systemic analgesia in thoracic trauma. For patients with rib fractures, catheter-based blocks provide better analgesic control and a lower incidence of pulmonary complications compared with opioids alone [94]. Fascial plane blocks also offer the advantage of reducing in hospital opioid use, with a more stable haemodynamic profile [95–97].

Several regional approaches are available and should be chosen based on the clinical picture. The serratus anterior block is recommended in patients with unilateral anterior or lateral rib fractures. This block is also especially useful in patients unable to sit or roll, for instance in patients with concomitant spinal injuries or those living with obesity [98]. The erector spinae plane block is recommended for posterior rib fractures. For posterior or lateral rib fractures with poor response to erector spinae plane or serratus anterior blocks, a paravertebral block or thoracic epidural is recommended [98]. Paravertebral blocks offer excellent analgesia and are less likely than extrathoracic fascial plane blocks to require rescue epidural intervention [99]. For bilateral rib fractures, a thoracic epidural or any of the above-mentioned techniques performed bilaterally is recommended. Thoracic epidural blocks provide excellent analgesia; however, they are falling out of favour due to their possible complication profile, as well as difficulty in insertion especially in patients unable to be positioned appropriately [98, 100]. Erector spinae plane blocks are technically easier to perform with a better safety profile [101] and may be used in patients with coagulopathy or vertebral fractures where paravertebral blocks are contra-indicated [101]. The serratus anterior and erector spinae plane blocks provide comparable analgesia to thoracic epidurals at 48 h post-intervention [95].

Patients with traumatic injuries are at a risk of developing acute respiratory distress syndrome due to direct and indirect injury. For example, damage-associated molecular patterns including histones are released into the circulation due to cellular death and injury after trauma and are associated with subsequent development of indirect lung injury [102]. Lung protective ventilation strategies are recommended in thoracic trauma to minimise the risk of secondary iatrogenic injury. This includes lower tidal volumes of 4–8 $ml.kg^{-1}$ of predicted body weight; adjusting positive end-expiratory pressure to avoid alveolar collapse at the end of expiration; and minimising plateau pressure to < 30 cmH_2O [103]. Bronchopleural fistulae may also develop as a sequela of thoracic trauma. In this case, ventilation strategies include encouragement of spontaneous breathing and lowering tidal volumes and positive end-expiratory pressure. Large leaks may require interventions such as one-lung ventilation using a double lumen tube or a bronchial blocker; in severe cases extracorporeal membrane oxygenation should be considered [103].

Fluid management is also key. Crystalloids should be used judiciously, due to the possibility of dilutional coagulopathy and/or development of pulmonary oedema in patients vulnerable to evolving lung injury. An individualised approach is recommended including the early use of blood products in patients who are hypotensive, guided by point-of-care viscoelastic tests. Vasopressors should be used with caution as hypotension in a trauma

© 2026 Association of Anaesthetists.

13652044, 0, Downloaded from https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.70284 by Hacettepe University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

setting is usually due to haemorrhagic shock and vasoconstriction could potentiate organ hypoperfusion [104].

Tracheobronchial injuries occur in up to 0.4% of patients with multiple traumatic injuries [105]. Careful airway management is required to avoid causing further damage. Direct visualisation of the airway is recommended to minimise the chance of worsening a tear or creating a false lumen and ideally in a spontaneously breathing, awake patient. Awake tracheostomy is the intervention of choice in this situation with careful consideration of fibreoptic-assisted tracheal intubation as a secondary option [106]. Human factors are important in all clinical settings, but this cohort of patients is particularly challenging, with complex multidisciplinary resuscitation and management required [107].

Special considerations

Older patients (aged ≥65 y) who sustain thoracic trauma have higher mortality rates (3.4% vs. 1.0%, $p < 0.001$) and a higher risk of complications (7.0% vs. 4.7%, $p < 0.001$) compared with younger patients [108]. Age-related changes of the thoracic cavity predispose older patients to more severe injuries, with a prolonged recovery course [109]. Chest wall compliance decreases due to calcification of rib articular surfaces and ossification of costal cartilage. Osteoporotic changes lead to decreased mobility of the chest wall, with a more brittle rib cage that is less pliable to traumatic insults. In addition, ventilation/perfusion mismatch occurs as alveoli collapse at lower lung volumes [110]. Immune-related changes leave older patients more susceptible to lung infections, complicating recovery [111].

Concomitant medications, such as anticoagulants, increases risk of occult or more severe bleeding [112]. Older patients may also have a blunted physiologic response to shock due to pre-existing cardiac illness or drugs such as beta-blockers, reflecting in seemingly `normal´ vital signs. Serial lactate and base deficit measurements are recommended to detect `occult shock´ [113]. Thorough triaging is required for older patients, with low threshold for activation of the trauma response team including low threshold for CT scan imaging [114]. Older patients with thoracic trauma benefit from early critical care management. This is especially applicable for patients with several risk factors such as high frailty scores; ≥ five rib fractures and the need for intercostal drain insertion; or poor spirometry function [115]. These patients may require lower threshold for ICU admission to reduce the risk of complications such as pneumonia; tracheal intubation; and death [116].

Discussion

Thoracic trauma may be classified by mechanism of injury (blunt or penetrating) or by the organs involved. Several structures may be injured, with each organ involved presenting unique challenges in terms of diagnosis and management. Management may be further complicated in the presence of other injuries as well as the physiologic and metabolic derangements as a consequence of the initial trauma. Surgical intervention is often necessary, but equally important is supportive management, including optimum pain control; airway and ventilatory support; fluid regulation; mobilisation; and management of surgical drains and dressings.

A key strength of this review is incorporation of multidisciplinary perspectives from anaesthetic; intensive care; general surgery; vascular surgery; orthopaedic and radiological specialties. However, there are several limitations. We did not develop a prospective structured protocol for this review, instead drawing on the broad experience of authors to inform our search strategy and reporting. This approach may increase the potential for selective citation, but we reasoned that a review focussed on multidisciplinary perspectives would be a useful addition to the existing literature base. Furthermore, we have not sought to critically appraise the included literature, instead giving an overview of current clinical practice, favouring inclusion of relevant guidelines where these exist.

As highlighted by our review, major thoracic trauma is highly complex and requires close multidisciplinary collaboration to achieve optimal patient outcomes. In such complex systems, with interdependent care pathways, we recommend that implementation science approaches be applied to close the gap between research discovery and real-world practice. We suggest that optimal sustainable care pathways are co-created through close collaboration between key stakeholders (including patients) and subsequently tested within a cluster-randomised controlled trial to determine effectiveness with embedded health economic evaluation.

Author affiliations

Navniel Kaur,[1] Sophie Lewis,[2] Hollie Jenkinson,[3] Christopher Keegan,[4] Nikhil Misra,[3] Simon Neequaye,[5] Simon Scott,[6] Tim Astles,[2] and Ben Morton[2,7]

1. Department of Anaesthesia, Mbagathi County Referral Hospital, Nairobi, Kenya.

© 2026 Association of Anaesthetists.

13652044, 0, Downloaded from https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.70284 by Hacettepe University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

2. Critical Care Department, Aintree University Hospital, University Hospitals of Liverpool Group, Liverpool, UK.
3. Trauma and Emergency General Surgery Service, Aintree University Hospital, University Hospitals of Liverpool Group, Liverpool, UK.
4. Department of Radiology, Aintree University Hospital, University Hospitals of Liverpool Group, Liverpool, UK.
5. Liverpool Vascular & Endovascular Service, Aintree University Hospital, University Hospitals of Liverpool Group, Liverpool, UK.
6. Cheshire & Mersey Major Trauma Centre Collaborative, Aintree University Hospital, University Hospitals of Liverpool Group, Liverpool, UK.
7. Department of Clinical Sciences, Liverpool School of Tropical Medicine, Pembroke Place, Liverpool, UK.

Acknowledgements

BM is an Editor of *Anaesthesia*. No external funding or other competing interests declared.

References

1. Jain A, Waseem M. Chest trauma. StatPearls. 2025. https://www.ncbi.nlm.nih.gov/books/NBK482194 (accessed 21/10/2025).
2. Institute for Health Metrics and Evaluation. Global Burden of Disease 2023: Findings from the GBD 2023 Study. 2025. https://www.healthdata.org/research-analysis/library/global-burden-disease-2023-findings-gbd-2023-study (accessed 11/03/2026).
3. Lundin A, Akram SK, Berg L, Göransson KE, Enocson A. Thoracic injuries in trauma patients: epidemiology and its influence on mortality. *Scand J Trauma Resusc Emerg Med* 2022; **30**: 69. https://doi.org/10.1186/s13049-022-01058-6.
4. Singh AK, Prasad G, Mishra P. The impact of thoracic trauma on morbidity and outcomes: a six-year experience from a tertiary care level 1 center. *Cureus* 2024; **16**: e73580. https://doi.org/10.7759/cureus.73580.
5. Kibu OD, Nguefack-Tsague G, Maqungo S, et al. Factors associated with mortality in thoracic trauma patients in Cameroon. *Heliyon* 2024; **11**: e41499. https://doi.org/10.1016/j.heliyon.2024.e41499.
6. Recep D, Burak O, Kursad O, Semih H. Comprehensive analysis of 4205 patients with chest trauma: a 10-year experience. *Interact Cardiovasc Thorac Surg* 2009; **9**: 450–3. https://doi.org/10.1510/icvts.2009.206599.
7. Adal O, Tareke AA, Bogale EK, et al. Mortality of traumatic chest injury and its predictors across sub-Saharan Africa: systematic review and meta-analysis, 2024. *BMC Emerg Med* 2024; **24**: 32. https://doi.org/10.1186/s12873-024-00951-w.
8. Helsloot D, Fitzgerald M, Lefering R, Verelst S, Missant C. The first hour of trauma reception is critical for patients with major thoracic trauma: a retrospective analysis from the TraumaRegister DGU. *Eur J Anaesthesiol* 2023; **40**: 865–73. https://doi.org/10.1097/EJA.0000000000001834.
9. Association for the Advancement of Automotive Medicine. The Abbreviated Injury Scale 2005 - Update 2008. 2015. https://www.aaam.org/education-resource-center/public-position-statements/abbreviated-injury-scale-ais-position-statement (accessed 21/10/25).
10. Beshay M, Mertzlufft F, Kottkamp HW, Reymond M, Schmid RA, Branscheid D, Vordemvenne T. Analysis of risk factors in thoracic trauma patients with a comparison of a modern trauma centre: a mono-centre study. *World J Emerg Surg* 2020; **15**: 45. https://doi.org/10.1186/s13017-020-00324-1.
11. Bayer J, Lefering R, Reinhardt S, Kühle J, Zwingmann J, Südkamp NP, Hammer T. Thoracic trauma severity contributes to differences in intensive care therapy and mortality of severely injured patients: analysis based on the TraumaRegister DGU®. *World J Emerg Surg* 2017; **12**: 43. https://doi.org/10.1186/s13017-017-0154-1.
12. Koutsouroumpa C, Leivaditis V, Mulita F, et al. Lung parenchymal trauma biomechanics, mechanisms, and classification: a narrative review of the current knowledge. *Pol J Cardio-Thorac Surg* 2025; **22**: 100–11. https://doi.org/10.5114/kitp.2025.152221.
13. Liman ST, Kuzucu A, Tastepe AI, Ulasan GN, Topcu S. Chest injury due to blunt trauma. *Eur J Cardiothorac Surg* 2003; **23**: 374–8. https://doi.org/10.1016/s1010-7940(02)00813-8.
14. Hanafi M, Al-Sarraf N, Sharaf H, Abdelaziz A. Pattern and presentation of blunt chest trauma among different age groups. *Asian Cardiovasc Thorac Ann* 2011; **19**: 48–51. https://doi.org/10.1177/0218492310395954.
15. Cakmak G, Cansun F, Saracoglu A, T Saracoglu K. Airway management in penetrating thoracic trauma. *Anaesthesiol Intensive Ther* 2022; **54**: 253–61. https://doi.org/10.5114/ait.2022.118332.
16. Reeder RJ, Davis DM. Penetrating chest trauma: with careful assessment, rapid diagnosis, and relatively simple interventions, most patients with penetrating chest trauma will recover. *Am J Nurs* 2001; **101**: 15–8. https://doi.org/10.1097/00000446-200109001-00004.
17. Stefanopoulos PK, Mikros G, Pinialidis DE, Oikonomakis IN, Tsiatis NE, Janzon B. Wound ballistics of military rifle bullets: an update on controversial issues and associated misconceptions. *J Trauma Acute Care Surg* 2019; **87**: 690–8. https://doi.org/10.1097/TA.0000000000002290.
18. Dogrul BN, Kiliccalan I, Asci ES, Peker SC. Blunt trauma related chest wall and pulmonary injuries: an overview. *Chin J Traumatol* 2020; **23**: 125–38. https://doi.org/10.1016/j.cjtee.2020.04.003.
19. Zhang S, Xiao X, Wang J, et al. Epidemiological and clinical characteristics of road traffic crashes related thoracic traumas: analysis of 5095 hospitalized chest injury patients. *J Cardiothorac Surg* 2021; **16**: 220. https://doi.org/10.1186/s13019-021-01599-4.
20. Veysi VT, Nikolaou VS, Paliobeis C, Efstathopoulos N, Giannoudis PV. Prevalence of chest trauma, associated injuries and mortality: a level I trauma centre experience. *Int Orthop* 2009; **33**: 1425–33. https://doi.org/10.1007/s00264-009-0746-9.
21. Coccolini F, Cremonini C, Moore EE, et al. Thoracic trauma WSES-AAST guidelines. *World J Emerg Surg* 2025; **20**: 78. https://doi.org/10.1186/s13017-025-00651-1.
22. American College of Surgeons. *Advanced Trauma Life Support (ATLS®) Student Course Manual*. 11th edn. 2025. https://www.facs.org/quality-programs/trauma/education/advanced-trauma-life-support/atls-11 (accessed 11/03/2026).
23. European Trauma Course Organisation. *European Trauma Course: The Team Approach*. 4th edn. 2019. https://publications.europeantraumacourse.com/view/845868233 (accessed 11/03/2026).
24. Wiles MD. ATLS: archaic trauma life support? *Anaesthesia* 2015; **70**: 893–7. https://doi.org/10.1111/anae.13166.
25. NHS Commissioning Board. NHS Standard Contract for Major Trauma Service (All Ages). 2013. https://www.england.nhs.uk/wp-content/uploads/2014/04/d15-major-trauma-0414.pdf (accessed 20/10/25).

© 2026 Association of Anaesthetists.

13652044, 0, Downloaded from https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.70284 by Hacettepe University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

26. Royal College of Radiologists. Major adult trauma radiology guidance. 2024. https://www.rcr.ac.uk/media/mbzdxefx/rcr-major-adult-trauma-guidance-2024.pdf (accessed 20/10/25).
27. Van Vledder MG, Kwakernaak V, Hagenaars T, et al. Patterns of injury and outcomes in the elderly patient with rib fractures: a multicenter observational study. *Eur J Trauma Emerg Surg* 2019; **45**: 575–83. https://doi.org/10.1007/s00068-018-0969-9.
28. Bhavnagri SJ, Mohammed TLH. When and how to image a suspected broken rib. *Cleve Clin J Med* 2009; **76**: 309–14. https://doi.org/10.3949/ccjm.76a.08026.
29. Kim HW, Kim YT, Jou SS, Han JK, Bae WK. Traumatic sternal fracture: incidence, causes, and CT features. *J Korean Soc Radiol* 2015; **72**: 305–12. https://doi.org/10.3348/jksr.2015.72.5.305.
30. Sairanen JJ, Arponen O. Isolated sternal fracture after low-energy trauma in a geriatric patient: a case report. *Int J Emerg Med* 2022; **15**: 34. https://doi.org/10.1186/s12245-022-00437-1.
31. Sermonesi G, Bertelli R, Pieracci FM, et al. Surgical stabilization of rib fractures (SSRF): the WSES and CWIS position paper. *World J Emerg Surg* 2024; **19**: 33. https://doi.org/10.1186/s13017-024-00559-2.
32. Pieracci FM, Majercik S, Ali-Osman F, et al. Consensus statement: surgical stabilization of rib fractures rib fracture colloquium clinical practice guidelines. *Injury* 2017; **48**: 307–21. https://doi.org/10.1016/j.injury.2016.11.026.
33. Chest Wall Injury Society. Indications for surgical stabilization of rib fractures. 2020. https://cwisociety.org/wp-content/uploads/2020/05/CWIS-SSRF-Guideline-01102020.pdf (accessed 20/10/25).
34. Kong LW, Huang GB, Yi YF, Du DY. The Chinese consensus for surgical treatment of traumatic rib fractures 2021 (C-STTRF 2021). *Chin J Traumatol* 2021; **24**: 311–9. https://doi.org/10.1016/j.cjtee.2021.07.012.
35. Pieracci FM, Leasia K, Bauman Z, et al. A multicenter, prospective, controlled clinical trial of surgical stabilization of rib fractures in patients with severe, nonflail fracture patterns. *J Trauma Acute Care Surg* 2020; **88**: 249–57. https://doi.org/10.1097/TA.0000000000002559.
36. de Campos JRM, White TW. Chest wall stabilization in trauma patients: why, when, and how? *J Thorac Dis* 2018; **10**: S951–62. https://doi.org/10.21037/jtd.2018.04.69.
37. Fokin AA, Hus N, Wycech J, Rodriguez E, Puente I. Surgical stabilization of rib fractures: indications, techniques, and pitfalls. *JBJS Essent Surg Tech* 2020; **10**: e0032. https://doi.org/10.2106/JBJS.ST.19.00032.
38. Marasco S, Saxena P. Surgical rib fixation—technical aspects. *Injury* 2015; **46**: 929–32. https://doi.org/10.1016/j.injury.2014.12.021.57.
39. Lewis B, Herr K, Hamlin S, et al. Imaging manifestations of chest trauma. *Radiographics* 2021; **41**: 1321–34. https://doi.org/10.1148/rg.2021210042.
40. Bader A, Rahman U, Morris M, et al. Pulmonary contusions in the elderly after blunt trauma: incidence and outcomes. *J Surg Res* 2018; **230**: 110–6. https://doi.org/10.1016/j.jss.2018.04.049.
41. Finkbeiner R, Krinner S, Langenbach A, Besendörfer M, Schulz-Drost S. Age distribution and concomitant injuries in pulmonary contusion: an analysis based on routine data. *Thorac Cardiovasc Surg* 2018; **66**: 678–85. https://doi.org/10.1055/s-0038-1667323.
42. Rendeki S, Molnár TF. Pulmonary contusion. *J Thorac Dis* 2019; **11**(Suppl 2): S141–51. https://doi.org/10.21037/jtd.2018.11.53.
43. Cohn SM, Dubose JJ. Pulmonary contusion: an update on recent advances in clinical management. *World J Surg* 2010; **34**: 1959–70. https://doi.org/10.1007/s00268-010-0599-9.
44. Ketai L, Primack SL. Thoracic Trauma. In: Hodler J, Kubik-Huch RA, von Schulthess GK, eds. *Diseases of the Chest, Breast, Heart and Vessels 2019–2022*. Cham: Springer, 2019: 139–54. https://doi.org/10.1007/978-3-030-11149-6.
45. Lee N, Kim S, Seo S, et al. Prediction of respiratory complications by quantifying lung contusion volume using chest computed tomography in patients with chest trauma. *Sci Rep* 2023; **13**: 6387. https://doi.org/10.1038/s41598-023-33275-z.
46. Miller D, Mansour K. Blunt traumatic lung injuries. *Thorac Surg Clin* 2007; **17**: 57–61. https://doi.org/10.1016/j.thorsurg.2007.03.017.
47. Daurat A, Millet I, Roustan JP, et al. Thoracic trauma severity score on admission allows determination of the risk of delayed ARDS in trauma patients with pulmonary contusion. *Injury* 2016; **47**: 147–53. https://doi.org/10.1016/j.injury.2015.08.031.
48. Miller LA. Chest wall, lung, and pleural space trauma. *Radiol Clin North Am* 2006; **44**: 213–24. https://doi.org/10.1016/j.rcl.2005.10.006.
49. Zugazaga A, Stachno MA, García A, et al. Pulmonary artery pseudoaneurysms: endovascular management after adequate imaging diagnosis. *Eur Radiol* 2021; **31**: 6480–8. https://doi.org/10.1007/s00330-021-07819-8.
50. Mahmood I, Younis B, Ahmed K, et al. Occult pneumothorax in patients presenting with blunt chest trauma: an observational analysis. *Qatar Med J* 2020; **2020**: 10. https://doi.org/10.5339/qmj.2020.10.
51. Bou Zein Eddine S, Boyle K, Dodgion C, et al. Observing pneumothoraces: the 35-millimeter rule is safe for both blunt and penetrating chest trauma. *J Trauma Acute Care Surg* 2019; **86**: 557–64. https://doi.org/10.1097/TA.0000000000002192.
52. Barrios C, Tran T, Malinoski D, et al. Successful management of occult pneumothorax without tube thoracostomy despite positive pressure ventilation. *Am Surg* 2008; **74**: 958–61.
53. Emergency Medicine Cases. Episode 174 – Traumatic Pneumothorax – Is Less More? 2022. https://emergencymedicinecases.com/wp-content/uploads/2022/09/Episode-174-Traumatic-Pneumothorax.pdf (accessed 3/1/26).
54. Clements T, Sirois M, Parry N, et al. OPTICC: a multicentre trial of occult pneumothoraces subjected to mechanical ventilation. *Am J Surg* 2021; **221**: 1252–8.
55. Kulvatunyou N, Bauman ZM, Zein Edine SB, et al. The small (14 Fr) percutaneous catheter versus large (28–32 Fr) chest tube for traumatic hemothorax: a multicenter randomized clinical trial. *J Trauma Acute Care Surg* 2021; **91**: 809–13. https://doi.org/10.1097/ta.0000000000003180.
56. Kulvatunyou N, Erickson L, Vijayasekaran A, et al. Randomized clinical trial of pigtail catheter versus chest tube in injured patients with uncomplicated traumatic pneumothorax. *Br J Surg* 2014; **101**: 17–22. https://doi.org/10.1002/bjs.9377.
57. Freixinet Gilart J, Ramírez Gil ME, Gallardo Valera G, Moreno CP. Chest trauma. *Arch Bronconeumol* 2011; **47**(Suppl 3): 9–14. https://doi.org/10.1016/s0300-2896(11)70023-0.
58. Gonzalez G, Robert C, Petit L, Biais M, Carrié C. May the initial CT scan predict delayed hemothorax in blunt chest trauma patients? *Eur J Trauma Emerg Surg* 2021; **47**: 71–8. https://doi.org/10.1007/s00068-020-01391-4.
59. Teyi T, James JD, Kumar V, et al. No role of antibiotics in patients with chest trauma requiring inter-costal drain: a pilot randomized controlled trial. *Eur J Trauma Emerg Surg* 2023; **49**: 1113–20. https://doi.org/10.1007/s00068-022-02163-y.

© 2026 Association of Anaesthetists.

13652044, 0, Downloaded from https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.70284 by Hacettepe University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

60. Pumarejo Gomez L, Tran VH. Hemothorax. StatPearls. 2025. https://www.ncbi.nlm.nih.gov/books/NBK538219 (accessed 20/10/25).
61. Boersma W, Stigt J, Smit H. Treatment of haemothorax. *Respir Med* 2010; **104**: 1583–7. https://doi.org/10.1016/j.rmed.2010.08.006.
62. Alborzi Z, Zangouri V, Paydar S, et al. Diagnosing myocardial contusion after blunt chest trauma. *J Tehran Heart Cent* 2016; **11**: 49–54.
63. Van Lieshout EMM, Verhofstad MHJ, Van Silfhout DJT, et al. Diagnostic approach for myocardial contusion: retrospective evaluation and literature review. *Eur J Trauma Emerg Surg* 2021; **47**: 1259–72. https://doi.org/10.1007/s00068-020-01305-4.
64. Latif R, Clifford S, Ghafghazi S, et al. Echocardiography and management for cardiac trauma. *J Cardiothorac Vasc Anesth* 2022; **36**: 3265–77. https://doi.org/10.1053/j.jvca.2022.02.010.
65. Singh S, Heard M, Pester JM, et al. Blunt cardiac injury. StatPearls. 2025. https://www.ncbi.nlm.nih.gov/books/NBK532267 (accessed 20/10/25).
66. Nazir S, Sharma R. Echocardiographic evaluation of cardiac trauma. Anesthesia Key2019. https://aneskey.com/echocardiographic-evaluation-of-cardiac-trauma (accessed 3/1/26).
67. Adler Y, Ristić AD, Imazio M, et al. Cardiac tamponade. *Nat Rev Dis Primers* 2023; **9**: 36. https://doi.org/10.1038/s41572-023-00446-1.
68. Pendleton AC, Leichtle SW. Cardiac tamponade from blunt trauma. *Am Surg* 2020; **88**: 1319–21. https://doi.org/10.1177/0003134820942170.
69. Stashko E, Meer JM. Cardiac Tamponade. StatPearls. 2025. https://www.ncbi.nlm.nih.gov/books/NBK431090 (accessed 19/10/25).
70. Assaf M, Abdalla A, Shaltout AE, et al. Pericardial tamponade in trauma: a systematic review of diagnosis, emergency management, and surgical outcomes. *Cureus* 2025; **17**: e91921. https://doi.org/10.7759/cureus.91921.
71. Lashin H, Helal N, Mahran M, Sharif A. Pattern, characterization and outcomes of blunt and penetrating cardiothoracic injuries: a medicolegal comparative study. *Legal Med* 2025; **76**: 102654. https://doi.org/10.1016/j.legalmed.2025.102654.
72. Hromalik LR Jr, Wall MJ Jr, Mattox KL, Tsai PI. Penetrating cardiac injury: a narrative review. *Media* 2023; **7**: 15. https://doi.org/10.21037/med-22-18.
73. Lee C, Jebbia M, Morchi R, Grigorian A, Nahmias J. Cardiac trauma: a review of penetrating and blunt cardiac injuries. *Am Surg* 2024; **91**: 423–33. https://doi.org/10.1177/00031348241307400.
74. Paulich S, Lockey D. Resuscitative thoracotomy. *BJA Educ* 2020; **20**: 242–8. https://doi.org/10.1016/j.bjae.2020.03.005.
75. Lott C, Truhlář A, Alfonzo A, et al. European resuscitation council guidelines 2021: cardiac arrest in special circumstances. *Resuscitation* 2021; **161**: 152–219. https://doi.org/10.1016/j.resuscitation.2021.08.012.
76. Elhajj A, Dakessian A, Bachir R, El Sayed M. Factors associated with survival after emergency department thoracotomy for adult trauma patients in the United States. *J Emerg Med* 2020; **59**: 169–77. https://doi.org/10.1016/j.jemermed.2020.05.005.
77. Perkins ZB, Greenhalgh R, ter Avest E, et al. Prehospital resuscitative thoracotomy for traumatic cardiac arrest. *JAMA Surg* 2025; **160**: 432–40. https://doi.org/10.1001/jamasurg.2024.7245.
78. Wahlgren CM, Aylwin C, Davenport RA, et al. European Society for Vascular Surgery (ESVS) 2025 clinical practice guidelines on the management of vascular trauma. *Eur J Vasc Endovasc Surg* 2025; **69**: 179–237. https://doi.org/10.1016/j.ejvs.2024.12.018.
79. Osgood M, Heck J, Rellinger E, et al. Natural history of grade I–II blunt traumatic aortic injury. *J Vasc Surg* 2014; **59**: 334–41. https://doi.org/10.1016/j.jvs.2013.09.007.
80. Slocum C, Chiba H, Emigh B, Tam B, Schellenberg M, Inaba K, Matsushima K. Nationwide analysis of penetrating thoracic aortic injury: injury patterns, management, and outcomes. *J Surg Res* 2023; **284**: 290–5. https://doi.org/10.1016/j.jss.2022.11.077.
81. Tang GL, Tehrani HY, Usman A, Katariya K, Otero C, Perez E, Eskandari MK. Reduced mortality, paraplegia, and stroke with stent graft repair of blunt aortic transections: a modern meta-analysis. *J Vasc Surg* 2008; **47**: 671–5. https://doi.org/10.1016/j.jvs.2007.08.031.
82. Mill V, Wahlgren CM, Dias N, et al. Long term outcomes of endovascular repair for blunt traumatic aortic injury: a twenty year multicentre follow up study. *Eur J Vasc Endovasc Surg* 2025; **69**: 382–90. https://doi.org/10.1016/j.ejvs.2024.10.048.
83. Demetriades D, Velmahos GC, Scalea TM, et al. Operative repair or endovascular stent graft in blunt traumatic thoracic aortic injuries: results of an American Association for the Surgery of Trauma Multicenter Study. *J Trauma* 2008; **64**: 561–70. https://doi.org/10.1097/TA.0b013e3181641bb3.
84. Abdellatif W, Chow B, Hamid S, Khorshed D, Khosa F, Nicolaou S, Murray N. Unravelling the mysteries of traumatic diaphragmatic injury: an up-to-date review. *Can Assoc Radiol J* 2020; **71**: 313–21. https://doi.org/10.1177/0846537120905133.
85. Simon LV, Lopez RA, Burns B. Diaphragm Rupture. 2025. https://www.ncbi.nlm.nih.gov/books/NBK470214 (accessed 20/10/25).
86. Dahal S, Koju R, Shrestha B, Karki T, Bade S. Spleen in the thorax: a case report on traumatic diaphragmatic rupture. *Int J Surg Case Rep* 2020; **77**: 664–7. https://doi.org/10.1016/j.ijscr.2020.11.018.
87. Cremonini C, Lewis MR, Jakob D, Benjamin ER, Chiarugi M, Demetriades D. Diagnosing penetrating diaphragmatic injuries: CT scan is valuable but not reliable. *Injury* 2022; **53**: 116–21. https://doi.org/10.1016/j.injury.2021.09.014.
88. Epailly J, Georges O, de Dominicis F. Traumatic diaphragmatic wound repair. *J Visc Surg* 2024; **161**: 262–6. https://doi.org/10.1016/j.jviscsurg.2024.06.006.
89. Battle C, Hutchings H, Lovett S, et al. Predicting outcomes after blunt chest wall trauma: development and external validation of a new prognostic model. *Crit Care* 2014; **18**: R98. https://doi.org/10.1186/cc13873.
90. Chapman BC, Herbert B, Rodil M, et al. RibScore: a novel radiographic score based on fracture pattern that predicts pneumonia, respiratory failure, and tracheostomy. *J Trauma Acute Care Surg* 2016; **80**: 95–101. https://doi.org/10.1097/TA.0000000000000907.
91. Pressley CM, Fry WR, Philp AS, Berry SD, Smith RS. Predicting outcome of patients with chest wall injury. *Am J Surg* 2012; **204**: 910–4. https://doi.org/10.1016/j.amjsurg.2012.05.015.
92. Easter A. Management of patients with multiple rib fractures. *Am J Crit Care* 2001; **10**: 320–7. https://doi.org/10.4037/ajcc2001.10.5.320.
93. Maxwell C, Mion L, Dietrich M, et al. Hospitalised injured older adults: clinical utility of a rib fracture scoring system. *J Trauma Nurs* 2012; **19**: 168–74. https://doi.org/10.1097/jtn.0b013e318261d201.
94. Peek J, Smeeing DPJ, Hietbrink F, Houwert RM, Marsman M, de Jong MB. Comparison of analgesic interventions for traumatic rib fractures: a systematic review and meta-analysis. *Eur J Trauma Emerg Surg* 2019; **45**: 597–622. https://doi.org/10.1007/s00068-018-0918-7.

© 2026 Association of Anaesthetists.

13652044, 0, Downloaded from https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.70284 by Hacettepe University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

95. Lin BH, Huang HM, Lin SF. Efficacy and safety of serratus anterior plane block and erector spinae plane block for rib fracture pain: a systematic review and meta-analysis. *Clin J Pain* 2026; **42**: e1334. https://doi.org/10.1097/AJP.0000000000001334.
96. Niazi AU, Solish M, Moorthy A, Niazi F, Abate AH, Devion C, Choi S. Use of fascial plane blocks for traumatic rib fractures: a scoping review. *Reg Anesth Pain Med* 2026; **51**: 414–28. https://doi.org/10.1136/rapm-2024-106366.
97. Partyka C, Asha S, Berry M, et al. Serratus anterior plane blocks for early rib fracture pain management: the SABRE randomized clinical trial. *JAMA Surg* 2024; **159**: 810–7. https://doi.org/10.1001/jamasurg.2024.0969.
98. Williams A, Bigham C, Marchbank A. Anaesthetic and surgical management of rib fractures. *BJA Educ* 2020; **20**: 332–40. https://doi.org/10.1016/j.bjae.2020.06.001.
99. Ramin S, Charbit J, Airoldi F, et al. Continuous posterior extrathoracic fascial plane block versus continuous paravertebral block for pain management in patients with multiple rib fractures: a randomised controlled noninferiority trial. *Br J Anaesth* 2026; **136**: 677–86. https://doi.org/10.1016/j.bja.2025.09.058.
100. Mitchell J, Couvreur C, Forget P. Recent advances in perioperative analgesia in thoracic surgery: a narrative review. *J Clin Med* 2024; **14**: 38. https://doi.org/10.3390/jcm14010038.
101. Adhikary SD, Liu WM, Fuller E, Cruz-Eng H, Chin KJ. The effect of erector spinae plane block on respiratory and analgesic outcomes in multiple rib fractures: a retrospective cohort study. *Anaesthesia* 2019; **74**: 585–93. https://doi.org/10.1111/anae.14579.
102. Abrams ST, Zhang N, Manson J, et al. Circulating histones are mediators of trauma-associated lung injury. *Am J Respir Crit Care Med* 2013; **187**: 160–9. https://doi.org/10.1164/rccm.201206-1037OC.
103. Arora S, Singh PM, Trikha A. Ventilatory strategies in trauma patients. *J Emerg Trauma Shock* 2014; **7**: 25–31. https://doi.org/10.4103/0974-2700.125635.
104. Richards JE, Harris T, Dünser MW, Bouzat P, Gauss T. Vasopressors in trauma: a never event? *Anesth Analg* 2021; **133**: 68–79. https://doi.org/10.1213/ANE.0000000000005552.
105. Schibilsky D, Driessen A, White WJ, et al. Traumatic tracheobronchial injuries: incidence and outcome of 136.389 patients derived from the DGU trauma register. *Sci Rep* 2020; **10**: 20555. https://doi.org/10.1038/s41598-020-77613-x.
106. Mercer SJ, Jones CP, Bridge M, Clitheroe E, Morton B, Groom P. Systematic review of the anaesthetic management of non-iatrogenic acute adult airway trauma. *Br J Anaesth* 2016; **117**: i49–59. https://doi.org/10.1093/bja/aew193.
107. Jones CPL, Fawker-Corbett J, Groom P, Morton B, Lister C, Mercer SJ. Human factors in preventing complications in anaesthesia: a systematic review. *Anaesthesia* 2018; **73**(Suppl 1): 12–24. https://doi.org/10.1111/anae.14136.
108. Panossian VS, Lagazzi E, Rafaqat W, et al. Outcomes of severe isolated blunt chest trauma in young and geriatric patients. *Eur J Trauma Emerg Surg* 2024; **50**: 2471–80. https://doi.org/10.1007/s00068-024-02611-x.
109. Qaqish T, Coleman J, Katlic M. Thoracic trauma in the elderly. In: Rodriguez A, Barraco RD, Ivatury RR, eds. *Geriatric Trauma and Acute Care Surgery*. Cham: Springer, 2017: 157–66. https://doi.org/10.1007/978-3-319-57403-5_17.
110. Davies GA, Bolton CE. Age related changes in the respiratory system. In: Fillit HM, Rockwood K, Woodhouse K, eds. *Brocklehurst's Textbook of Geriatric Medicine and Gerontology*, 7th edn. Amsterdam: Elsevier, 2010: 97–100.
111. Gomez CR, Boehmer ED, Kovacs EJ. The aging innate immune system. *Curr Opin Immunol* 2005; **17**: 457–62. https://doi.org/10.1016/j.coi.2005.07.013.
112. Christie DB 3rd, Nowack T, Drahos A, Ashley DW. Geriatric chest wall injury: is it time for a new sense of urgency? *J Thorac Dis* 2019; **11**(Suppl 8): S1029–33. https://doi.org/10.21037/jtd.2018.12.16.
113. De Simone B, Chouillard E, Podda M, et al. The 2023 WSES guidelines on the management of trauma in elderly and frail patients. *World J Emerg Surg* 2024; **19**: 18. https://doi.org/10.1186/s13017-024-00537-8.
114. Yang AR, Lundy M, Otaibi B, et al. Triage of the injured older adult: a narrative review of nationwide hospital practices. *Am J Surg* 2025; **249**: 116615. https://doi.org/10.1016/j.amjsurg.2025.116615.
115. Fields AC, Castillo-Angeles M, Bryan MR, et al. Rib fracture injury guidelines for patients over the age of 65: should these patients routinely be admitted to the intensive care unit? *J Surg Res* 2025; **315**: 443–50. https://doi.org/10.1016/j.jss.2025.09.051.
116. Bowman JA, Nuño M, Jurkovich GJ, Utter GH. Association of hospital-level intensive care unit use and outcomes in older patients with isolated rib fractures. *JAMA Netw Open* 2020; **3**: e2026500. https://doi.org/10.1001/jamanetworkopen.2020.26500.

© 2026 Association of Anaesthetists.

13652044, 0, Downloaded from https://associationofanaesthetists-publications.onlinelibrary.wiley.com/doi/10.1111/anae.70294 by Hacettepe University, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License